Product Introduction

The ZDH1060 is a high-gain, high-efficiency power amplifier based on InGaP technology. It is suitable for a wide voltage supply of 5~9V and operates at a frequency of DC-1000MHz. Under a typical 7.4V supply, it can deliver up to 6.5W of output power in the 400MHz~470MHz range with an efficiency of 55%. Input and output are 50Ω matched.

The ZDH1060 uses a standard QFN8x8-56PIN package and features low cost and high reliability.

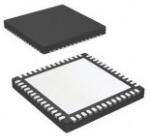
Typical application scenarios

- Mobile Radio Amplifier
- Walkie-Talkie Amplifier
- Wireless Data Transmission (FSK, AFSK, OOK, GFSK etc.)

Absolute Maximum Ratings

Parameter	Value
Storage Temperature	-65°C~+150°C
Operating Temperature	-55°C~+105°C
Supply Voltage (Vcc)	9.0V
Supply Current (Icc)	5.0A
Input Power (RFIN)	15.0dBm

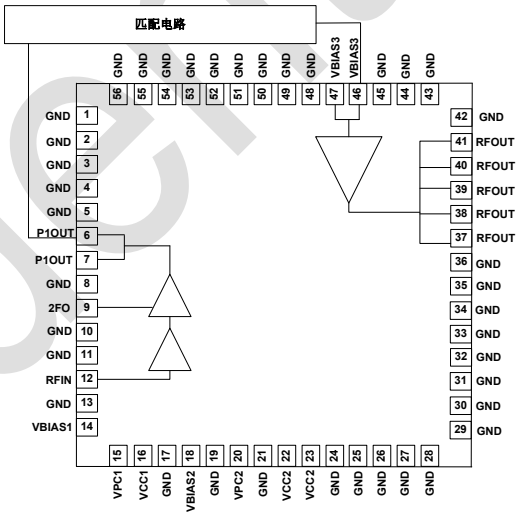
Features



- 5V~9V single supply voltage, typical current 1600mA @ 7.4V
- Typical gain: 37dB @ 440MHz
- Typical P1dB: 38dBm @ 440MHz
- Efficiency: >55% @ 440MHz
- Input and output 50Ω matched
- Green lead-free 56-pin QFN8x8 package

This product complies with all relevant regulations and is halogen-free

Pin diagram (Top View)



PIN No.	Pin Name	Description
6,7	P1OUT	Second-stage signal output
9	2FO	Matching settings
12	RFIN	Radio frequency signal input
14	VBIAS1	First-stage bias voltage
15	VPC1	First-stage drive voltage
16	VCC1	First-stage power supply voltage
18	VBIAS2	Second-stage bias voltage
20	VPC2	Second-stage drive voltage
22,23	VCC2	Second-stage power supply voltage
37,38,39,40,41	RFOUT	Radio frequency output
46,47	VBIAS3	Final-stage bias voltage
Other pins	GND	GND
57	EPAD	GND

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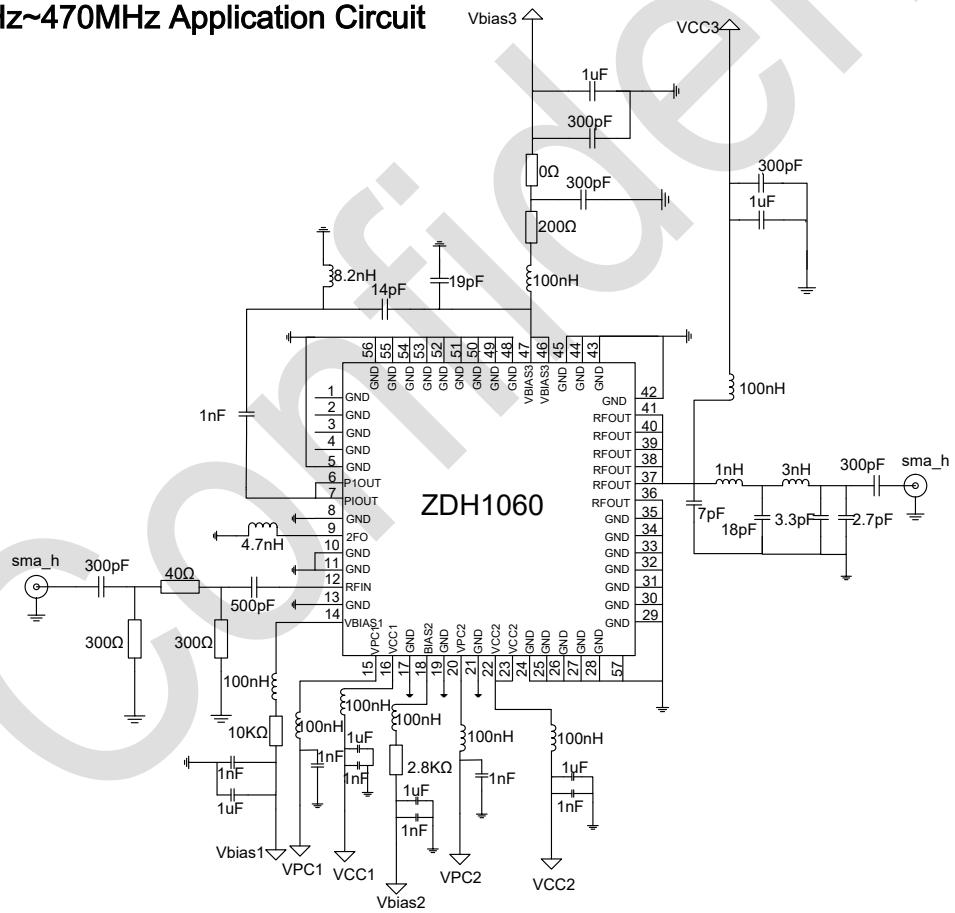
Typical application electrical parameters

Test conditions: VCC = +7.4 V, temperature = +25°C, using 400MHz~470 MHz application circuit, 50Ω test system.

Parameter	Minimum	Typical	Maximum	Unit	Condition
Frequency	400	440	470	MHz	-
Gain	-	37	-	dB	-
Gain Flatness	-	±0.3	-	dB	-
Input Return Loss (S11)	-	-9	-	dB	-
Output Return Loss (S22)	-	-18	-	dB	-
Input/Output Isolation (S12)	-	-45	-	dB	-
P1dB	-	38	-	dBm	Measured at 440MHz
Operating Voltage (VCC)	5	7.4	9	V	-
Operating Current (Icc)	-	1.6	-	A	Pout=6.5W
Quiescient Current (Icq)	-	1.2	-	A	-
Efficiency (η)	-	55	-	%	Pout=6.5W
Shutdown Current	-	8	-	uA	VBIAS1=VBIAS2=VBIAS3=VPC1=VPC2=0V

Typical Application:
400MHz~470MHz Application Circuit



Notes:

1. Refer to PCB material FR4, dielectric constant 4.6, board thickness 0.8mm, 1oz copper plating. See also Page 4 for recommended layout.
2. The bottom EPAD of the chip must have good contact with the PCB via vias to ensure proper heat dissipation.
3. In typical applications, Vbias1, Vbias2, Vbias3, VPC1, VPC2, VCC1, VCC2, and VCC3 are all set to 7.4V.
4. The L4 100nH inductor model is 1812SMS-101J.

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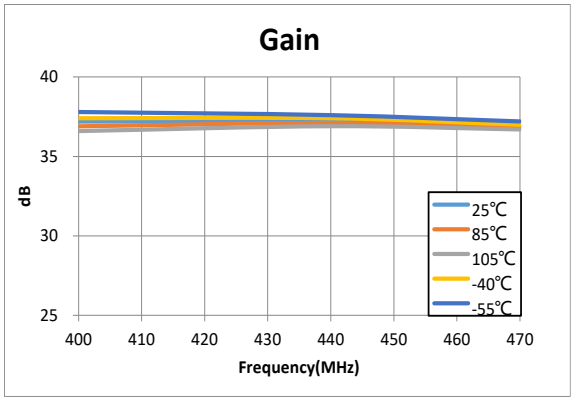
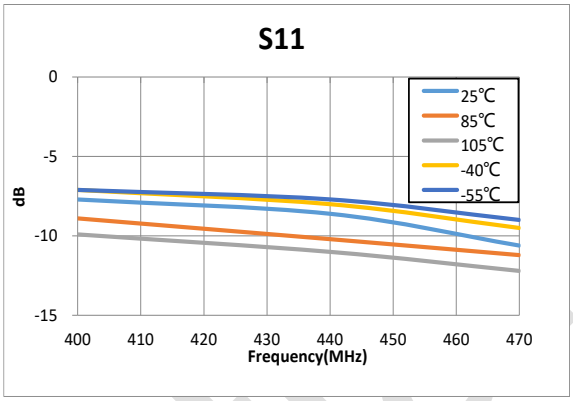
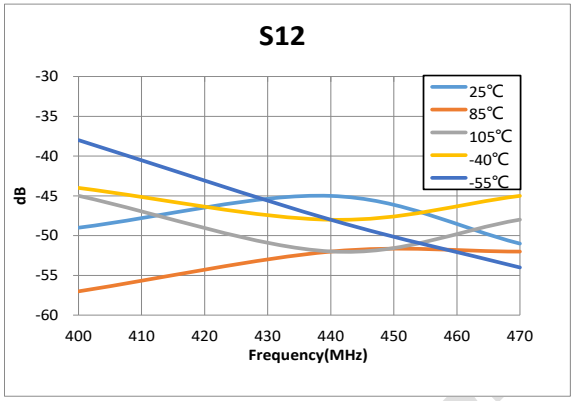
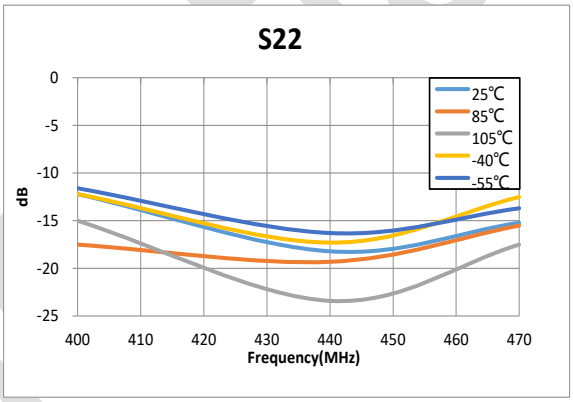
Typical performance curves

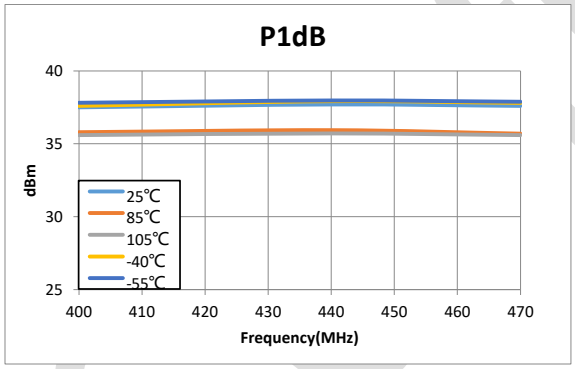
Test conditions: VCC = +7.4 V, using 400MHz~470 MHz application circuit, PIN = 0dBm, 50Ω test system.

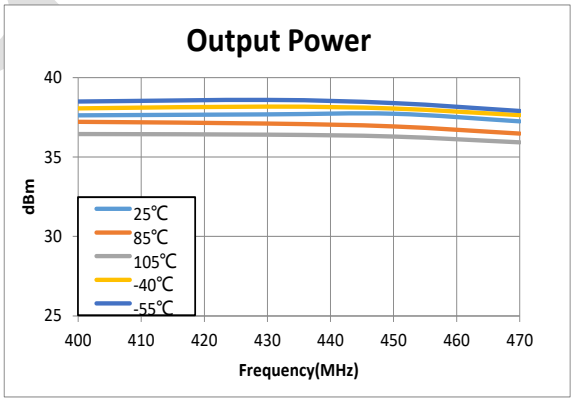


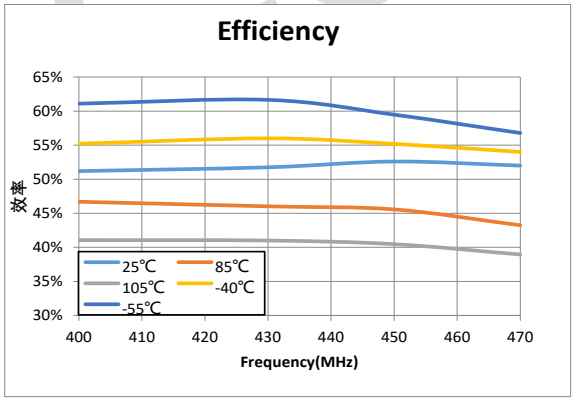




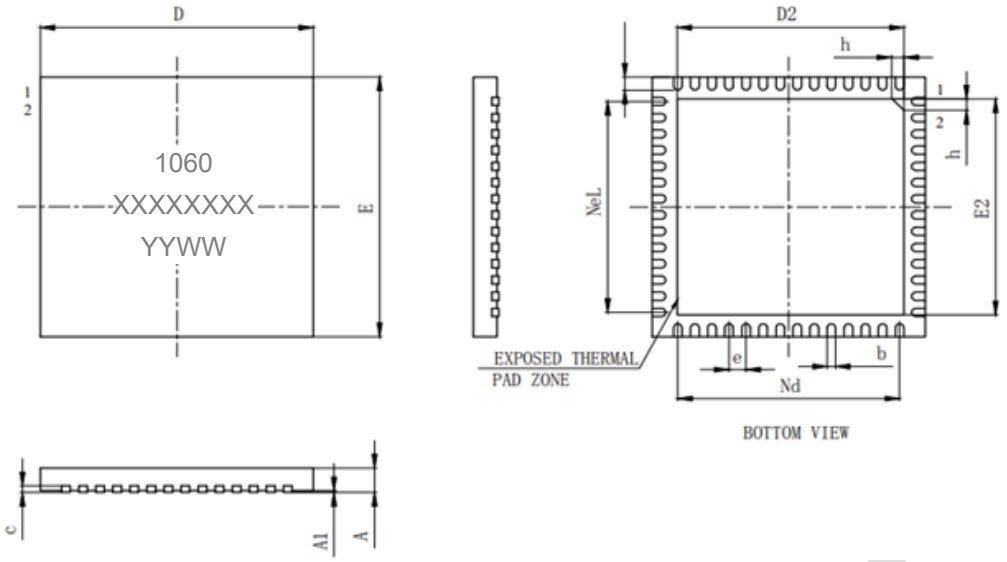








Package size diagram



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0. 70	0. 75	0. 80
	0. 80	0. 85	0. 90
A1	—	0. 02	0. 05
b	0. 18	0. 25	0. 30
c	0. 18	0. 20	0. 25
D	7. 90	8. 00	8. 10
D2	6. 55	6. 65	6. 75
e	0. 50BSC		
Ne	6. 50BSC		
Nd	6. 50BSC		
E	7. 90	8. 00	8. 10
E2	6. 55	6. 65	6. 75
L	0. 35	0. 40	0. 45
h	0. 30	0. 35	0. 40
载体尺寸 (mil)	270X270		

Notes:

1. Standard package thickness (A) is 0.75 mm.
2. Unit is mm.

Ordering Information

Model	Marking	Packaging
ZDH1060	1060	QFN8x8-56

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